

**COSC2129/1476**  
**Artificial Intelligence**

---

**Propositional Logic**

# Road Map for Today

---

- ❑ Revision of adversarial search
- ❑ Knowledge representation and reasoning
- ❑ Logic in general
  - Models and entailment
  - Soundness and completeness
- ❑ Propositional logic
  - Syntax and semantics
  - Model checking
  - Inference rules and theorem proving

# Search & Adversarial Search

---

- ❑ Uninformed search
- ❑ Informed search
- ❑ Games and adversarial search
  - Minimax search
  - Alpha-beta pruning search
  - Cutting off search
  - Heuristic evaluation function



# Knowledge and Reasoning

---

- ❑ Humans know things and what they know (knowledge) help them do things (reasoning).
- ❑ Intelligence of humans is achieved by the processes of reasoning that operates on internal representations of knowledge.
- ❑ It is also important for an artificial being, we call it **agent**, to achieve “high-level” intelligence.

A good chess program can defeat human masters but it does not know that a chess board can be used for playing checker as well. It cannot decide when to play a chess game.

The transition model for 8-puzzle problems, i.e., knowledge of what the actions do, can be used to predict the outcomes of actions but not to deduce that two tiles cannot occupy the same space.

# Knowledge and Reasoning (cont'd)

---

## □ Knowledge and reasoning

- enable agents to cope with complex environments.

[*E.g., how to schedule an around-world trip? Need the knowledge about “the world”.*]

- play a crucial role in dealing with partially observable environments.

[*Part of states could be hidden. E.g., how to avoid collisions in a busy shopping mall?*]

- enable agents to handle complicated tasks, e.g. understanding natural language.

[*E.g., “Mike opened the door, found a chair and sat on it”. It refers to door or chair?*]

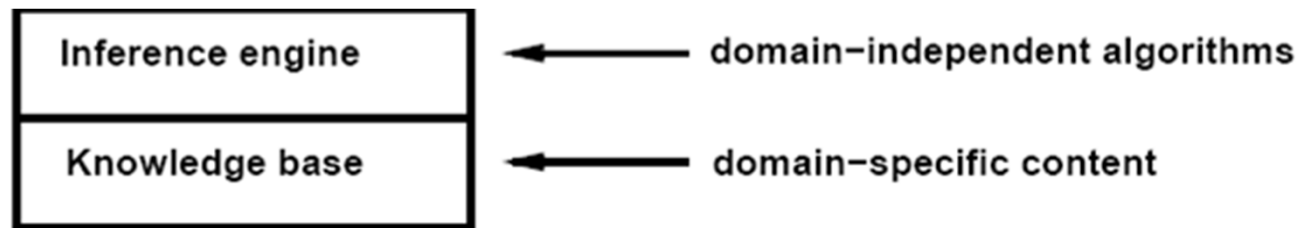
- provide more flexibility in problem solving.

[*Decision based on updated knowledge. E.g., how to choose a good text book for a course?* ]

# Knowledge and Reasoning (cont'd)

---

To be intelligent, an agent needs a **knowledge base** to store knowledge and an **inference engine** to do the reasoning.



An agent must be able to:

- represent knowledge of the world, e.g., states, actions, etc.
- incorporate new percepts.
- update internal representations of the world.
- deduce hidden properties of the world.
- deduce appropriate actions.

How to represent knowledge and perform inference?

# Knowledge Representation

---

## ❑ Desirable properties of a knowledge representation (KR) scheme:

- **Expressive**: must be able to represent as much as possible about the world.
- **Precise**: must be clear and unambiguous.  
E.g., the man saw the boy in the park with the telescope. Whose telescope?
- **Regular**: must be a clear mapping between the knowledge and its representation.
- **Adaptable**: must be able to add new information or delete invalid information.
- **Suitable for reasoning**: new knowledge can be inferred.
- **Computationally efficient**: the scheme is able to be implemented.
- **Qualitative**: it can represent qualitative knowledge.  
E.g., how to represent “you can only put block A on block B if B is clear”?
- **Meta-level reasoning**: can represent knowledge about knowledge.
- ...

# Wumpus World

## □ Rules of the environment

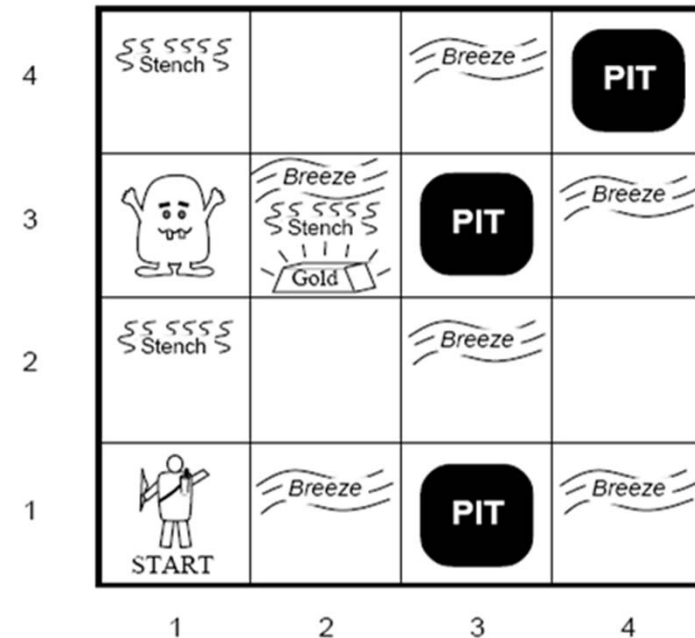
- Rooms adjacent to wumpus are smelly.
- Rooms adjacent to pit are breezy.
- The room containing gold is glittering.
- A bump is perceived if hitting into a wall.
- Scream spreads anywhere if wumpus is killed.
- Shooting kills wumpus if you are facing it.
- Shooting uses up the only arrow.
- Grabbing picks up gold if in same square.

## □ Percepts (sensors): Stench, Breeze, Glitter, Bump, Scream

## □ Actions: Left turn, Right turn, Forward, Grab, Shoot, Climb

## □ Partially observable

## □ Main challenge: Initial ignorance of the environmental configuration



- a cave containing 4x4 grid of rooms
- a wumpus lives in one of these room.
- some rooms contain bottomless pits.
- one room has a heap of gold.
- locations of wumpus and gold are random.
- each room may have a pit with prob. of 0.2.
- an agent starts from room [1, 1], facing right.
- room [1,1] is both the entrance and exit
- room [1,1] is free of wumpus, pit and gold.



# Wumpus World (cont'd)

---

Only partially observable, so reasoning is needed!

How does the agent reason?

- Represent the states of the world
- Determine the rules of the world, e.g., rooms next to the wumpus stink.

Then repeatedly:

- Incorporate percepts as inputs from sensors
- Update knowledge of the world
- Deduce "hidden" properties of the world
- Determine actions

# Wumpus World (cont'd)

---

Often considered as a knowledge base and an inference engine.

Knowledge base (domain-dependent)

"I am in room [3,2]."

"If the wumpus is in any adjacent room, I will sense a stench."

Inference engine (domain-independent)

"If A and  $A \Rightarrow B$ , then B."

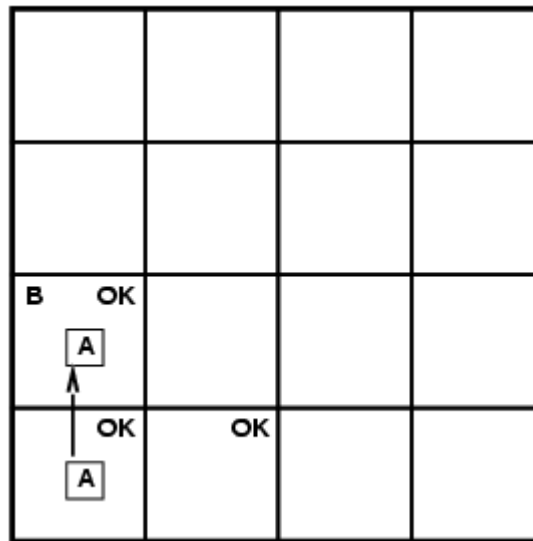
# Exploring a Wumpus World

---

|                    |    |  |  |
|--------------------|----|--|--|
|                    |    |  |  |
|                    |    |  |  |
| OK                 |    |  |  |
| OK<br><div>A</div> | OK |  |  |

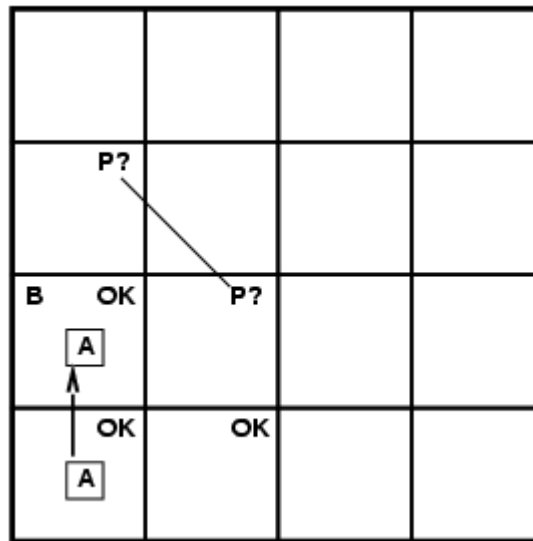
# Exploring a Wumpus World (cont'd)

---



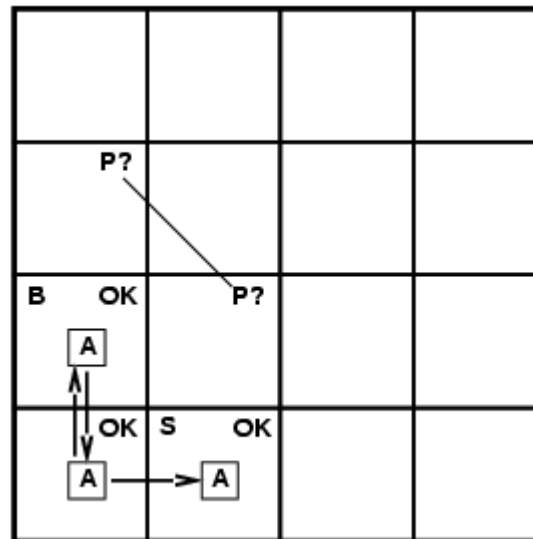
# Exploring a Wumpus World (cont'd)

---



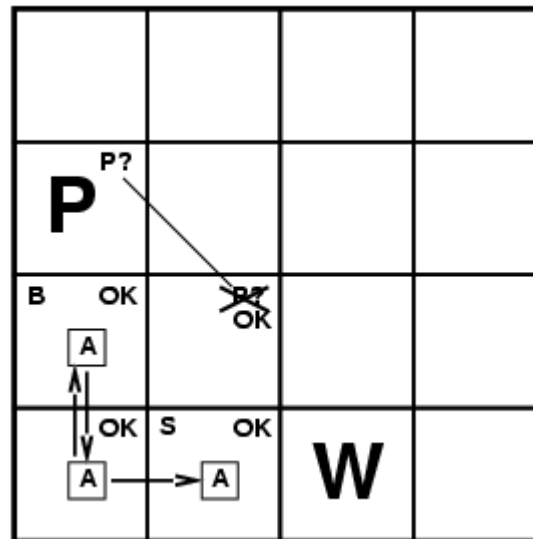
# Exploring a Wumpus World (cont'd)

---



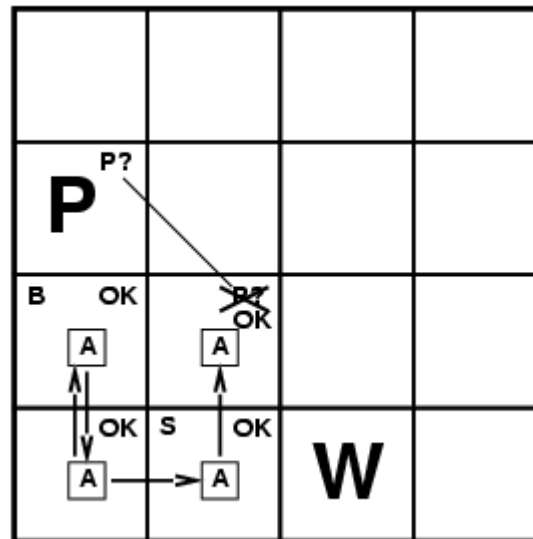
# Exploring a Wumpus World (cont'd)

---



# Exploring a Wumpus World (cont'd)

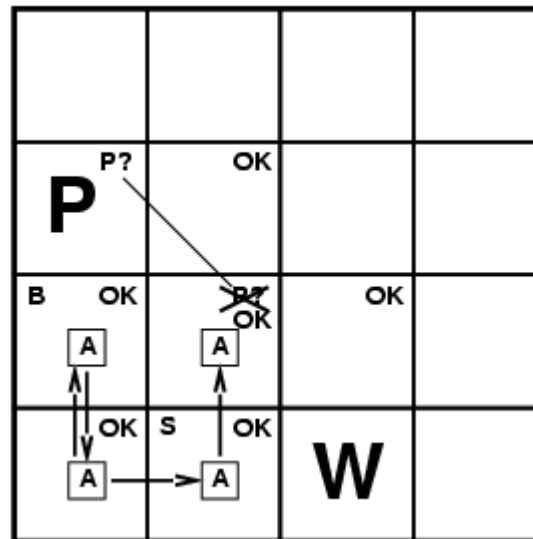
---





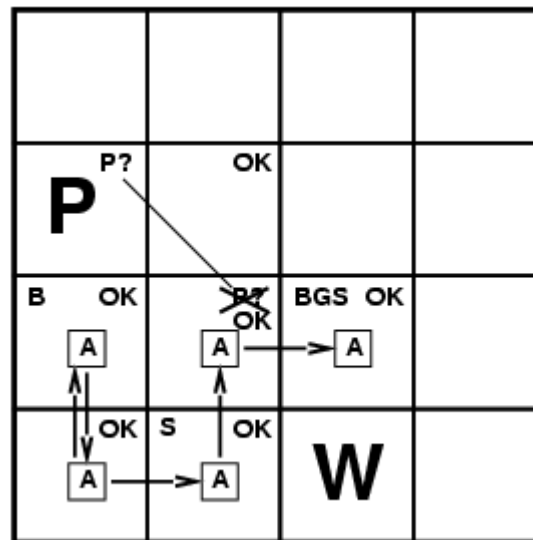
# Exploring a Wumpus World (cont'd)

---



# Exploring a Wumpus World (cont'd)

---



# Logic in General

---

- ❑ **Logic** is a set of formal languages for representing information such that conclusions can be drawn.
- ❑ **Syntax** specifies the well-formed sentences in the language.
- ❑ **Semantics** define the “meaning” of sentences, i.e., the truth of a sentence with respect to each possible world.
- ❑ Examples of the language of arithmetic:
  - $x+2 \geq y$  is a sentence;  $x^2+y > \{\}$  is not a sentence
  - $x+2 \geq y$  is true iff the number  $x+2$  is no less than the number  $y$
  - $x+2 \geq y$  is true in a world where  $x = 7, y = 1$
  - $x+2 \geq y$  is false in a world where  $x = 0, y = 6$

# Entailment

---

- Logical entailment means that one follows logically from another:

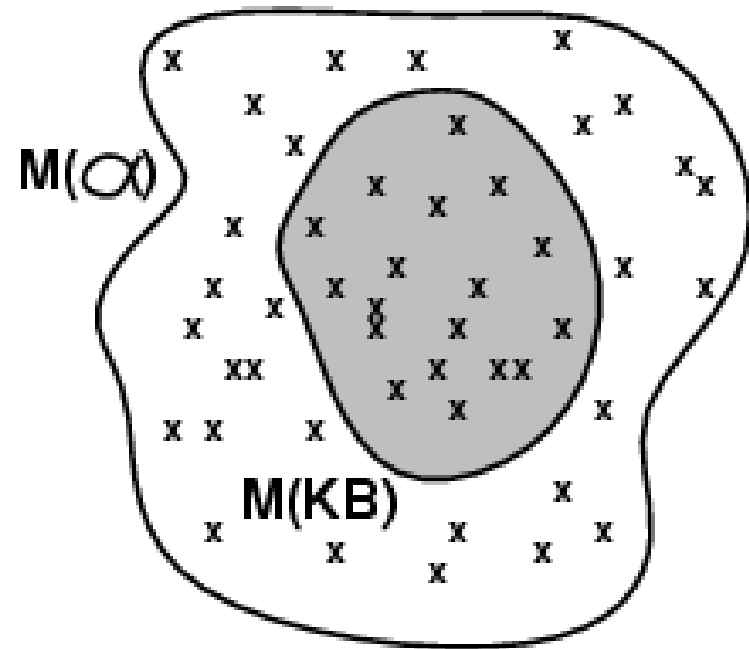
$$KB \models \alpha$$

- Knowledge base  $KB$  entails sentence  $\alpha$  **if and only if**  $\alpha$  is true in all worlds where  $KB$  is true.
  - $KB$  containing “the Giants won” and “the Reds won” entails “Either the Giants won or the Reds won”.
  - “ $x+y = 4$ ” entails “ $4 = x+y$ ”.
- Entailment is a relationship between sentences (syntax), which is based on semantics.

# Models

---

- ❑ Logicians typically think in terms of **models**, which are formally structured worlds with respect to which truth can be evaluated.  
In other words: Assignment of (T/F) values to each of the symbols.
- ❑ We say  $m$  is a **model of (or satisfies)** a sentence  $\alpha$  if  $\alpha$  is true in  $m$ .
- ❑  $M(\alpha)$  is the set of all models of  $\alpha$ .
- ❑  $KB \models \alpha$  iff  $M(KB) \subseteq M(\alpha)$ 
  - $KB$  = “the Giants won and the Reds won” and  $\alpha$  = “the Giants won”.



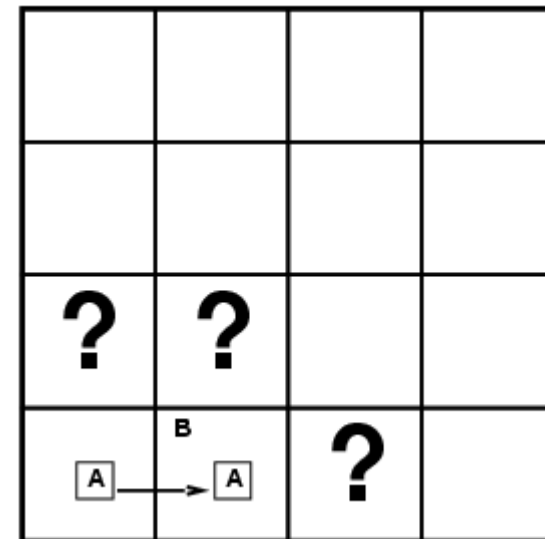
# Entailment in the Wumpus World

---

Situation after detecting nothing  
in [1,1], moving to [2, 1],  
and detecting breeze in [2,1]

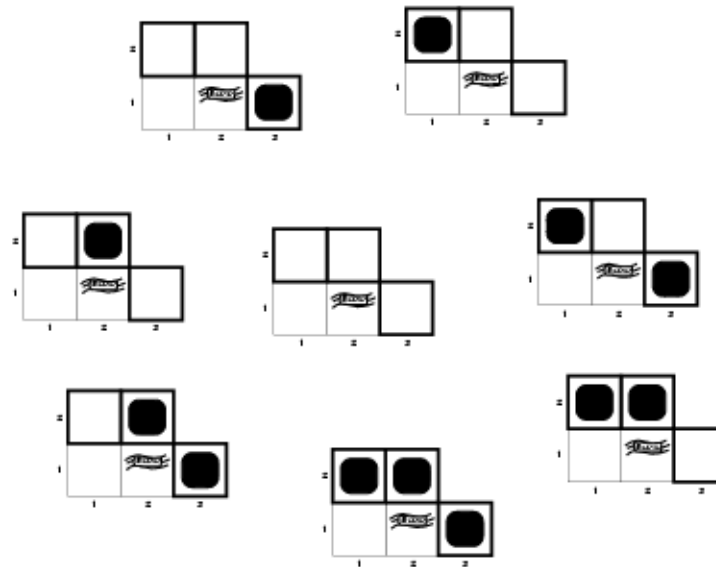
Consider possible models for ?s  
assuming only pits

3 Boolean choices  $\Rightarrow$  8 possible  
models



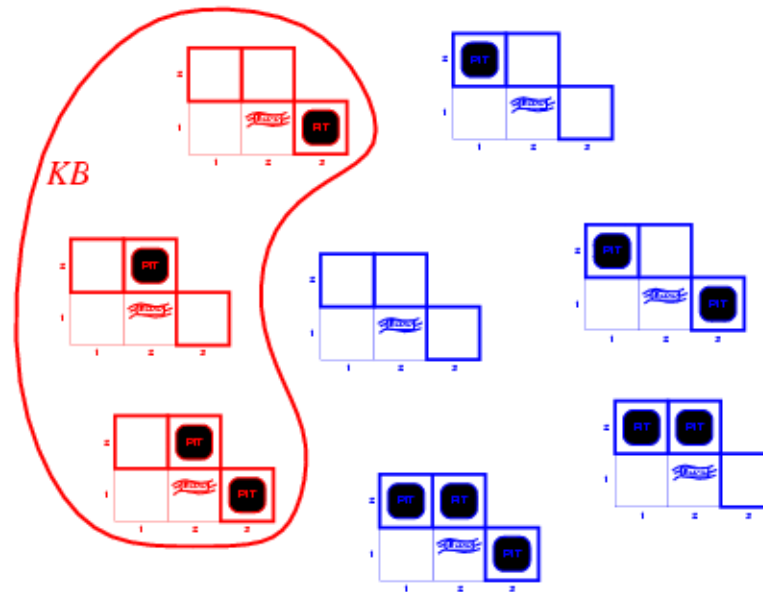
# Wumpus Models

---



# Wumpus Models (cont'd)

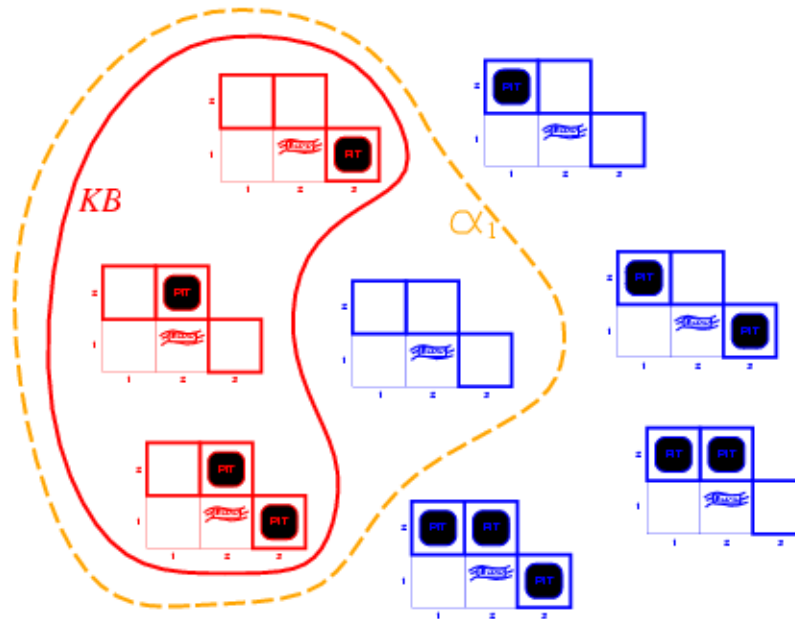
---



*KB* = the rules of the wumpus world + percepts



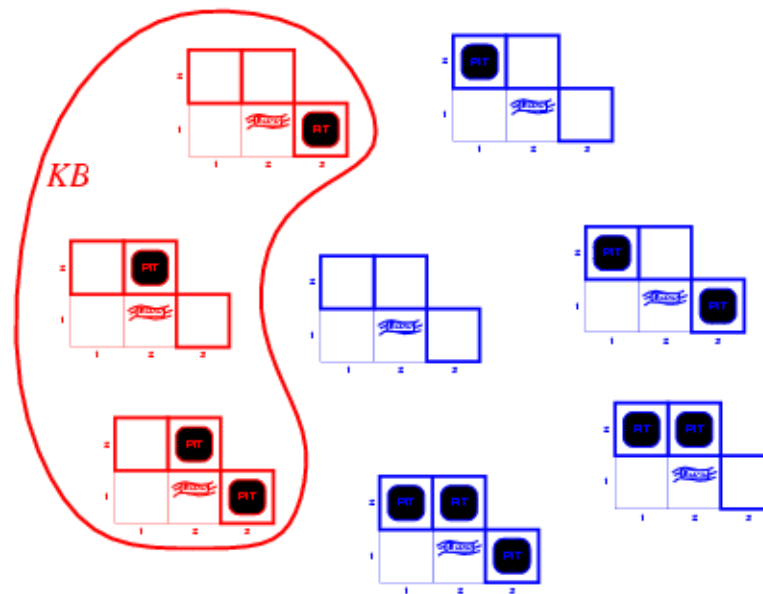
# Wumpus Models (cont'd)



- $KB$  = the rules of the wumpus world + percepts
- $\alpha_1$  = “[1, 2] is safe”,  $KB \models \alpha_1$ , proved by **model checking**

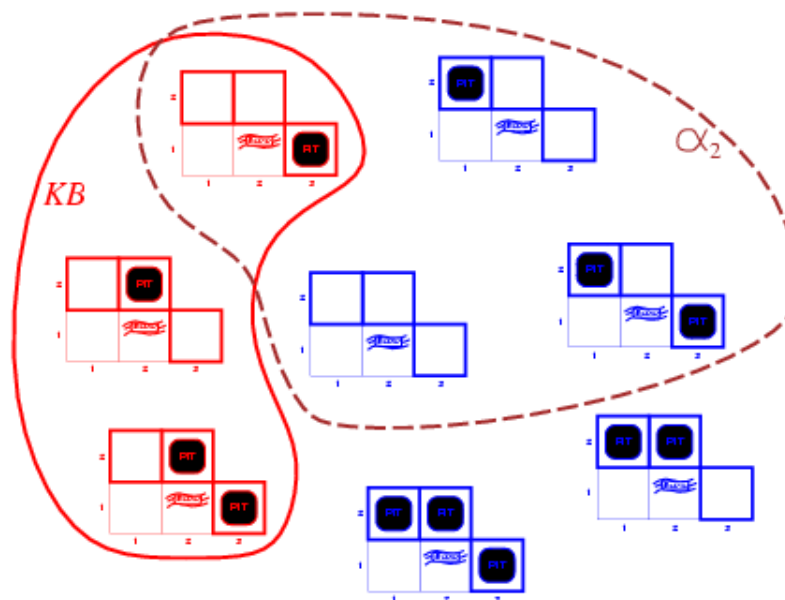
# Wumpus Models (cont'd)

---



*KB* = the rules of the wumpus world + percepts

# Wumpus Models (cont'd)



- ❑  $KB$  = the rules of the wumpus world + percepts
- ❑  $\alpha_2$  = “[2,2] is safe”,  $KB \models \alpha_2$

# Inference

---

- ❑  $KB \vdash_i \alpha$  denotes  $\alpha$  is derived from  $KB$  by an inference algorithm  $i$ .
  - Consequences of  $KB$  are a haystack;  $\alpha$  is a needle.
  - Entailment = needle in haystack; inference = finding it.
- ❑ **Soundness:**  $i$  is sound if  
whenever  $KB \vdash_i \alpha$ , it is also true that  $KB \models \alpha$
- ❑ **Completeness:**  $i$  is complete if  
whenever  $KB \models \alpha$ , it is also true that  $KB \vdash_i \alpha$
- ❑ Preview: we will define a logic (first-order logic) which is expressive enough to say almost anything of interest, and for which there exists a sound and complete inference procedure.
- ❑ The procedure will answer any question whose answer follows from what is known by the  $KB$ .

# Propositional Logic: Syntax

---

Propositional logic is the simplest logic!

Basic elements:

-- Propositional symbols (atomic sentences)

--  $P, Q, R, \dots$  (start with an uppercase letter)

--  $T$  (always true),  $F$  (always-false)

-- Connectives  $\neg, \wedge, \vee, \Rightarrow, \Leftrightarrow$

Complex sentences are constructed from simpler ones using connectives and parentheses.

$P_{3,1} \vee P_{3,2}$  : "There is a pit in  $[3,1]$  or  $[3,2]$ "

$P_{3,1} \wedge P_{3,2}$  : "There is a pit in  $[3,1]$  and  $[3,2]$ "

# Propositional Logic: Syntax (cont'd)

---

$T$  and  $F$  are atomic sentences

$P, Q, R, \dots$  are atomic sentences

If  $P$  is a sentence, so is  $(P)$  and  $[P]$

If  $P$  is a sentence, so is  $\neg P$

If  $P$  and  $Q$  are sentences, so are

$P \wedge Q$  (conjunction)

$P \vee Q$  (disjunction)

$P \Rightarrow Q$  (implication) a.k.a. rules or if-then statement

$P \Leftrightarrow Q$  (biconditional)

A **literal** is a atomic sentence either positive ( $P$ ) or negative ( $\neg P$ ).

# Propositional Logic: Syntax (cont'd)

---

Wff (well formed formula, i.e., syntactically ok):

$P$

$P \wedge Q$

$P \vee Q$

$(P \vee Q) \Rightarrow R$

$((P \wedge Q) \Rightarrow (\neg R \vee (S \wedge P)))$

**Note:** Parentheses must be used to avoid ambiguity.

Non-wff ("syntactic junk"):

$P \neg$

$P \wedge \neg$

$\wedge P Q$

$\Rightarrow P$

# Propositional Logic: Semantics

---

- A model specifies the truth value (true/false) for every proposition symbol

e.g.,  $P_{1,2}$        $P_{2,2}$        $P_{3,1}$   
false      true      false

With 3 symbols, 8 possible models can be enumerated automatically.

- Recursive evaluation: **atomic sentences** and **sentences formed with five connectives**.

- Rules for computing the truth value with respect to a model  $m$ :

-- T and F are true and false in every model, respectively

-- Other atomic sentences must be directly specified in  $m$

--  $\neg S$  is true iff  $S$  is false

--  $S_1 \wedge S_2$  is true iff  $S_1$  is true **and**  $S_2$  is true

--  $S_1 \vee S_2$  is true iff  $S_1$  is true **or**  $S_2$  is true

--  $S_1 \Rightarrow S_2$  is true iff  $S_1$  is false **or**  $S_2$  is true

is false iff  $S_1$  is true **and**  $S_2$  is false

--  $S_1 \Leftrightarrow S_2$  is true iff  $S_1 \Rightarrow S_2$  is true **and**  $S_2 \Rightarrow S_1$  is true

- Simple recursive process evaluates an arbitrary sentence, e.g.,

$$\neg P_{1,2} \wedge (P_{2,2} \vee P_{3,1}) = \text{true} \wedge (\text{true} \vee \text{false}) = \text{true} \wedge \text{true} = \text{true}$$



# Truth Tables

---

- Rules can be expressed with **truth tables**, specifying the truth values of sentences for each assignment of truth values to their components.
- Truth tables for 5 connectives:

| $P$          | $Q$          | $\neg P$     | $P \wedge Q$ | $P \vee Q$   | $P \Rightarrow Q$ | $P \Leftrightarrow Q$ |
|--------------|--------------|--------------|--------------|--------------|-------------------|-----------------------|
| <i>false</i> | <i>false</i> | <i>true</i>  | <i>false</i> | <i>false</i> | <i>true</i>       | <i>true</i>           |
| <i>false</i> | <i>true</i>  | <i>true</i>  | <i>false</i> | <i>true</i>  | <i>true</i>       | <i>false</i>          |
| <i>true</i>  | <i>false</i> | <i>false</i> | <i>false</i> | <i>true</i>  | <i>false</i>      | <i>false</i>          |
| <i>true</i>  | <i>true</i>  | <i>false</i> | <i>true</i>  | <i>true</i>  | <i>true</i>       | <i>true</i>           |

# Implication

---

The truth value of implication “ $\Rightarrow$ ” is somewhat count-intuitive.

Let  $P$  refers to “He studied hard.”,  $Q$  refers to “He passed the exam.”.

So  $P \Rightarrow Q$  means “He studied hard implies he passed the exam.”.

- True when  $P$  true  $Q$  true – *he studied hard so he passed the exam.*
- True when  $P$  false  $Q$  false – *he didn't study hard so he didn't pass the exam.*
- False when  $P$  true  $Q$  false – *he studied hard but he didn't pass the exam.*
- **True** when  $P$  **false**  $Q$  **true** – *he didn't study hard but he did pass the exam.*

The last one seems bizarre. It actually says if he passed the exam, then the implication is true even if the premise/antecedent is false.

“*Dogs love cats implies cats hate dogs*” odd but logically true if cats hate dogs is always true.

**Question:** what is the truth value of  $P \Rightarrow P$ ?

# Propositional Logic in Wumpus World

---

Let  $P_{i,j}$  be true if there is a pit in  $[i, j]$ .

Let  $B_{i,j}$  be true if it is breezy in  $[i, j]$ .

- |                                                                   |                                         |
|-------------------------------------------------------------------|-----------------------------------------|
| R1: $\neg P_{1,1}$                                                | There is no pit in $[1, 1]$ .           |
| R2: $B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$              | Pits cause breezes in adjacent squares. |
| R3: $B_{2,1} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1})$ | Pits cause breezes in adjacent squares. |
| R4: $\neg B_{1,1}$                                                | There is no breeze in $[1, 1]$          |
| R5: $B_{2,1}$                                                     | There is a breeze in $[2, 1]$ .         |

Based on these given information, we can *infer*  $\alpha$ : there is no pit in  $[1, 2]$ .

# Truth Tables for Inference

---

| $B_{1,1}$    | $B_{2,1}$    | $P_{1,1}$    | $P_{1,2}$    | $P_{2,1}$    | $P_{2,2}$    | $P_{3,1}$    | $KB$               | $\alpha_1$         |
|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------------|--------------------|
| <i>false</i> | <i>false</i> | <i>false</i> | <i>false</i> | <i>false</i> | <i>false</i> | <i>false</i> | <i>false</i>       | <i>true</i>        |
| <i>false</i> | <i>false</i> | <i>false</i> | <i>false</i> | <i>false</i> | <i>false</i> | <i>true</i>  | <i>false</i>       | <i>true</i>        |
| $\vdots$     | $\vdots$     | $\vdots$     | $\vdots$     | $\vdots$     | $\vdots$     | $\vdots$     | $\vdots$           | $\vdots$           |
| <i>false</i> | <i>true</i>  | <i>false</i> | <i>false</i> | <i>false</i> | <i>false</i> | <i>false</i> | <i>false</i>       | <i>true</i>        |
| <i>false</i> | <i>true</i>  | <i>false</i> | <i>false</i> | <i>false</i> | <i>false</i> | <i>true</i>  | <u><i>true</i></u> | <u><i>true</i></u> |
| <i>false</i> | <i>true</i>  | <i>false</i> | <i>false</i> | <i>false</i> | <i>true</i>  | <i>false</i> | <u><i>true</i></u> | <u><i>true</i></u> |
| <i>false</i> | <i>true</i>  | <i>false</i> | <i>false</i> | <i>false</i> | <i>true</i>  | <i>true</i>  | <u><i>true</i></u> | <u><i>true</i></u> |
| <i>false</i> | <i>true</i>  | <i>false</i> | <i>false</i> | <i>true</i>  | <i>false</i> | <i>false</i> | <i>false</i>       | <i>true</i>        |
| $\vdots$     | $\vdots$     | $\vdots$     | $\vdots$     | $\vdots$     | $\vdots$     | $\vdots$     | $\vdots$           | $\vdots$           |
| <i>true</i>  | <i>true</i>  | <i>true</i>  | <i>true</i>  | <i>true</i>  | <i>true</i>  | <i>true</i>  | <i>false</i>       | <i>false</i>       |

Enumerating rows via different assignments to symbols).

If  $KB$  is true for a row, then checking  $\alpha$  for that row.

# Model-Checking

---

- ❑ Inference by enumerating all models (via truth tables)
- ❑ Sound and complete
- ❑ For  $n$  proposition symbols, time complexity:  $O(2^n)$ , and space complexity:  $O(n)$

# Equivalence

---

Two sentences are logically equivalent if and only if (iff) they are true in the same models (they have the same truth value for same input). It is expressed as:

$$\alpha \equiv \beta \text{ iff } \alpha \models \beta \text{ and } \beta \models \alpha$$

Some standard logical equivalences:

$$\begin{aligned}(\alpha \wedge \beta) &\equiv (\beta \wedge \alpha) && \text{commutativity of } \wedge \\(\alpha \vee \beta) &\equiv (\beta \vee \alpha) && \text{commutativity of } \vee \\((\alpha \wedge \beta) \wedge \gamma) &\equiv (\alpha \wedge (\beta \wedge \gamma)) && \text{associativity of } \wedge \\((\alpha \vee \beta) \vee \gamma) &\equiv (\alpha \vee (\beta \vee \gamma)) && \text{associativity of } \vee \\\neg(\neg\alpha) &\equiv \alpha && \text{double-negation elimination} \\(\alpha \Rightarrow \beta) &\equiv (\neg\beta \Rightarrow \neg\alpha) && \text{contraposition} \\(\alpha \Rightarrow \beta) &\equiv (\neg\alpha \vee \beta) && \text{implication elimination} \\(\alpha \Leftrightarrow \beta) &\equiv ((\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)) && \text{biconditional elimination} \\\neg(\alpha \wedge \beta) &\equiv (\neg\alpha \vee \neg\beta) && \text{de Morgan} \\\neg(\alpha \vee \beta) &\equiv (\neg\alpha \wedge \neg\beta) && \text{de Morgan} \\(\alpha \wedge (\beta \vee \gamma)) &\equiv ((\alpha \wedge \beta) \vee (\alpha \wedge \gamma)) && \text{distributivity of } \wedge \text{ over } \vee \\(\alpha \vee (\beta \wedge \gamma)) &\equiv ((\alpha \vee \beta) \wedge (\alpha \vee \gamma)) && \text{distributivity of } \vee \text{ over } \wedge\end{aligned}$$

$P \Rightarrow Q$  is logically equivalent to  $\neg P \vee Q$ .

# Validity

---

A sentence is **valid** if it is true in **all** models.

The following sentences are valid:

*True*

$P \vee \neg P$

$P \Rightarrow P$

$(P \wedge (P \Rightarrow Q)) \Rightarrow Q$

Valid sentences are also known as **tautologies** (necessarily true).

Validity can be used for inference - **Deduction Theorem**:

$KB \models \alpha$  if and only if  $(KB \Rightarrow \alpha)$  is valid

# Satisfiability

---

A sentence is **satisfiable** if it is true in **some** models e.g.

$$P \vee Q,$$

$$P$$

A sentence is **unsatisfiable** if it is true in **no** models e.g.

$$P \wedge \neg P$$

Satisfiability can be used for inference as:

**$KB \models \alpha$  if and only if  $(KB \wedge \neg \alpha)$  is unsatisfiable**

It is actually the standard mathematical proof technique of *reductio ad absurdum* (literally, “reduction to an absurd thing”).

It is also called **proof by refutation** or **proof by contradiction**.



# Inference

---

The **aim** of logical inference is to decide whether an entailment  $\alpha \models \beta$  is true.

To prove it, there are two main approaches:

-- by truth tables based enumeration ([model-checking approach](#))

The complexity of this approach is exponential.

-- by applying inference rules ([theorem proving approach](#))

**Inference rule**

$$\frac{\alpha_1, \alpha_2, \dots, \alpha_n}{\beta}$$

This general form of inference rules means “*if we have proof that each  $\alpha_i$  is true then this is extended to a proof that  $\beta$  is true.*”.

A chained inference (i.e. possibly using more than one rule) from  $\alpha$  to  $\beta$  ( $\alpha$  infers  $\beta$ ) is written:

$$\alpha \vdash \beta$$

# Inference Rules

---

One well known inference rule is called **Modus Ponens**:

$$\frac{\alpha \Rightarrow \beta, \alpha}{\beta}$$

It means that if  $\alpha \Rightarrow \beta$  and  $\alpha$  are given, then  $\beta$  can be inferred.

E.g., if  $(Hungry \wedge FoodisAvailable) \Rightarrow Eat$  and  $Hungry \wedge FoodisAvailable$  are given, then  $Eat$  can be inferred.

Another useful inference rule is **And-Elimination**:

$$\frac{\alpha \wedge \beta}{\alpha}$$

It means from a conjunction, any of the conjuncts can be inferred.

For example, from  $(Hungry \wedge FoodisAvailable)$ ,  $Hungry$  can be inferred.

# Inference Rules (cont'd)

---

Logical equivalence can be used as inferences rules, e.g., biconditional elimination yields two rules:

$$\frac{(\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)}{\alpha \Leftrightarrow \beta}$$

$$\frac{\alpha \Leftrightarrow \beta}{(\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)}$$

By using these rules, we can infer that there is no pit in [1, 2] (the wumpus world).

We already know that:

R1:  $\neg P_{1,1}$ ; R2:  $B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$ ; R3:  $B_{2,1} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1})$ ; R4:  $\neg B_{1,1}$ ; R5:  $B_{2,1}$

Apply biconditional elimination to sentence R2, we can refer that:

R6:  $(B_{1,1} \Rightarrow (P_{1,2} \vee P_{2,1})) \wedge ((P_{1,2} \vee P_{2,1}) \Rightarrow B_{1,1})$

# Inference Rules (cont'd)

---

Apply And-Elimination  $\frac{\alpha \wedge \beta}{\alpha}$  to sentence R6, we can refer that:

$$R7: ((P_{1,2} \vee P_{2,1}) \Rightarrow B_{1,1})$$

Apply contraposition to sentence R7, we get:

$$R8: (\neg B_{1,1} \Rightarrow \neg(P_{1,2} \vee P_{2,1}))$$

Apply Modus Ponens  $\frac{\alpha \Rightarrow \beta, \alpha}{\beta}$  to sentence R8 (with Sentence R4:  $\neg B_{1,1}$ ), we get

$$R9: \neg(P_{1,2} \vee P_{2,1})$$

Apply de Morgan's rule to sentence R9, we can refer that:

$$R10: \neg P_{1,2} \wedge \neg P_{2,1}$$

So there is not pit in both [1,2] and [2,1].

# Monotonicity

---

The previous inference process did not use sentences R1, R3, R5

R1:  $\neg P_{1,1}$  R2:  $B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$  R3:  $B_{2,1} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1})$  R4:  $\neg B_{1,1}$  R5:  $B_{2,1}$

because they don't have relevant propositions.

*Finding a proof can be highly efficient in practice because it can ignore irrelevant propositions, no matter how many of them there are.*

This property of logical systems follows a fundamental property: **Monotonicity**.

Monotonicity means for any sentences  $\alpha$  and  $\beta$

$$\text{if } KB \models \alpha \text{ then } KB \wedge \beta \models \alpha$$

Additional information  $\beta$  cannot invalidate any conclusion  $\alpha$  already inferred.

Conclusion follow inference rules regardless of what else is in the knowledge base.

# Resolution

---

- Resolution inference rule:

$$\frac{l_i \vee \dots \vee l_k, \quad m_1 \vee \dots \vee m_n}{l_i \vee \dots \vee l_{i-1} \vee l_{i+1} \vee \dots \vee l_k \vee m_1 \vee \dots \vee m_{j-1} \vee m_{j+1} \vee \dots \vee m_n}$$

where  $l_i$  and  $m_j$  are complementary literals.

$$\text{e.g., } \frac{P_{1,3} \vee P_{2,2}, \quad \neg P_{2,2}}{P_{1,3}}$$

- Resolution is sound and complete for propositional logic

# Conjunctive Normal Form

---

Resolution can work for **all propositional logical sentences**

**Trick:** need to convert to **conjunctive normal form (CNF)**

"Conjunction of disjunction of literals"

CNF:  $\text{Clause}_1 \wedge \dots \wedge \text{Clause}_n$

Clause:  $\text{Literal}_1 \vee \dots \vee \text{Literal}_m$

Literal:  $P \mid \neg P$

# Conversion to CNF

---

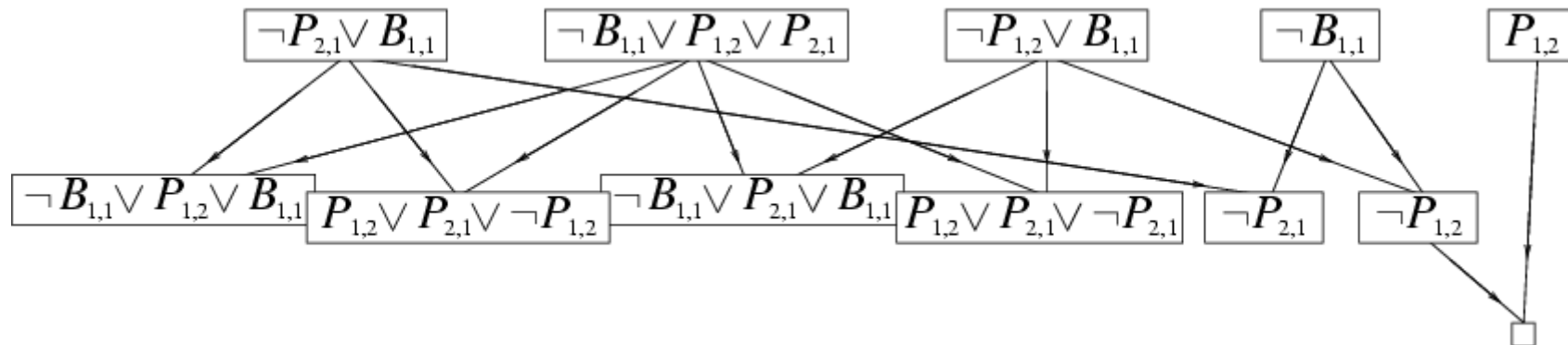
1. Replace  $P \Leftrightarrow Q$  with  $(P \Rightarrow Q) \wedge (Q \Rightarrow P)$
2. Replace  $P \Rightarrow Q$  with  $\neg P \vee Q$
3. Push  $\neg$  innermost
  - Replace  $\neg\neg P$  with  $P$
  - Replace  $\neg(P \vee Q)$  with  $\neg P \wedge \neg Q$
  - Replace  $\neg(P \wedge Q)$  with  $\neg P \vee \neg Q$
4. Distribute  $\vee$  over  $\wedge$ 
  - Replace  $P \vee (Q \wedge R)$  with  $(P \vee Q) \wedge (P \vee R)$



# Resolution example

---

$$KB = (B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})) \wedge \neg B_{1,1} \quad \alpha = \neg P_{1,2}$$



# Horn and Definite Clauses

---

**Horn clauses:** at most one positive literal

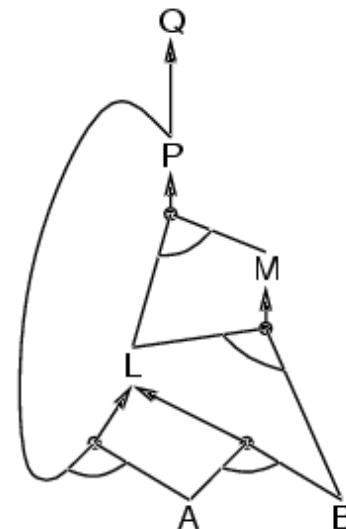
**Definite clauses:** exactly one positive literal

Resolution with two Horn clauses results in another Horn clause.

Deciding entailment with Horn clauses has linear time complexity.

Forward and backward chaining:

$P \Rightarrow Q$   
 $L \wedge M \Rightarrow P$   
 $B \wedge L \Rightarrow M$   
 $A \wedge P \Rightarrow L$   
 $A \wedge B \Rightarrow L$   
 $A$   
 $B$



# Conclusions

---

- ❑ Revision of adversarial search
- ❑ Knowledge representation and reasoning
- ❑ Logic in general
  - Models and entailment
  - Soundness and completeness
- ❑ Propositional logic
  - Syntax and semantics
  - Model checking
  - Inferences rules and theorem proving